16.2 A 28nm 53.8TOPS/W 8b Sparse Transformer Accelerator with In-Memory Butterfly Zero Skipper for Unstructured-Pruned NN and CIM-Based Local-Attention-Reusable Engine

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Transformer networks, from BERT, GPT to Alphafold, have demonstrated unprecedented advances in a variety of AI tasks. Fig. 16.2.1 shows the computing flow of self-attention – the fundamental operation in transformers. Queries (Q), keys (K) and values (V) are first obtained by multiplying inputs with 3 weight matrices. Afterward, scores that evaluate Q-K relevance are computed as scaled dot products and converted to probabilities through the softmax function. The probabilities are then multiplied by V, generating the final self-attention results. Transformer networks have led to an explosion in parameter counts, for example, 175B parameters for GPT-3. This demands significant growth in computing hardware and memory. Owing to expanding network sizes and corresponding power consumption, compute-in-memory (CIM) block-wise sparsity-aware architectures were proposed for matrix multiplication [1] and local attention [2] accelerators, where weight storage and compute are skipped for zero-value blocks. Yet, such structured sparsity is at the cost of notable accuracy loss [3]. Consequently, a challenge for CIM-based accelerators is in how to handle unstructured pruned NNs, while maintaining high efficiency. These unstructured patterns can be represented as: 1) irregularly distributed zero weights inside matrices, and 2) varied local attention spans for different attention heads.

To address the above challenges, this work presents an efficient transformer accelerator, featuring custom in-memory routing and computing macros for the aforementioned unstructured sparsity, and achieving 53.83TOPS/W energy efficiency and 0.85TOPS/mm^2 area efficiency in 28nm CMOS technology. The key contributions are: 1) a butterfly-network-based sparsity-aware feed-forward computing architecture, enabling irregularly distributed zero-weight skipping in macros, 2) a 32-8 in-memory sparse-to-dense routing array (IMPERA), extracting the inputs to be computed according to sparsity bit-masks, and 3) a digital CIM-based local-attention-reusable engine, facilitating a variable sparse attention span and QK sharing [4].

Figure 16.2.2 demonstrates the overall accelerator architecture, consisting of 4 transformer cores, a system controller and a 64KB input feature buffer. Each transformer core includes 16 feed-forward macros (FFM), an attention engine with 8 digital CIM macros (AM) and a post-processing engine (PPE) for computing log-softmax-based probability. Every FFM contains an IMPERA macro and a digital CIM macro. To fully accommodate self-attention in transformers, these blocks can be configured to three different modes. In the first mode, the FFMs compute Q and K sequentially, AMs perform $Q \bullet K^T$, and the PPE converts the AMs' results into probabilities in a pipeline. In the second mode, the FFMs compute V and AMs obtain the self-attention results of $P \bullet V$. The last mode is responsible for the final fully connected layer, where both the FFMs and AMs are set to complete matrix multiplication.

The butterfly-network-based sparsity-aware feed-forward zero-skipping scheme is shown in Fig. 16.2.3. Given sparse weight vectors, the proposed scheme first compresses them into dense vectors and binary-encoded bit-masks. The non-skipped input features need to be extracted and routed from the original inputs according to the bit-masks. The products are obtained by multiplying the two dense vectors. To maintain a routing array complexity of $O(N \bullet \log_2 1/S)$ (where N is the pruning window size and S is the non-zero ratio), a multi-stage cascaded butterfly network topology is utilized. To decode the bit-mask into the butterfly pass generation control bits, a prefix pop-count operation is first performed on the bit-mask. Then, the results are converted to control bits by a Left-Rotation and Complement-on-Warp (LRCW) operation. To compress the input vector, the last two stages of butterfly switches are realized with 4:1 MUXs. Experimental results show that accuracy is preserved until 75% of weights are pruned (S=0.25) for Enwik-8 and Text-8 – two common NLP datasets. In contrast with global pruning (N=512), a sub-bank irregular pruning (N=32) only results in 0.004/0.047 bits-per-character (BPC) loss on the above tasks, outperforming the 16×16 block-wise structured pruning by 0.088/0.126BPC. Therefore, we chose N=32 and S=0.25 to avoid an over-complicated butterfly implementation. To support a higher non-zero ratio, for example when S=0.5, the butterfly network selects 8 non-skipped features from 16 inputs. The unused inputs are gated.

Figure 16.2.4 illustrates the circuit design of the butterfly-based IMPERA macro. If the routing array were implemented by conventional digital VLSI methodology (synthesis-placement-routing), its power consumption will dominate the overall FFM. Therefore, an in-memory custom design method is applied to IMPERA, utilizing 10T-SRAM cells for the generated control-bit storage, and cascaded transmission gates (TG) for the butterfly network. The 10T cell employs a C^{2}MOS gate as a combinational read-out, eliminating sequential sense amplifiers. Simulation shows that the 10T cell dissipates 3.31× and 1.22× less power than 6T and 8T cells, respectively, in the routing array. To achieve a 32:8 compressor (25% non-zero ratio), the array contains a 3-stage butterfly network and single-stage 4:1 MUXs. The entire 32-entry butterfly network is split into 4 identical 8-entry sub-networks with different control bits. The MUXs select the outputs of sub-networks and generate the final extracted results. All switches are implemented by TGs and directly connected without buffers. Compared with conventional standard-cell digital routing arrays, the proposed design achieves 4.04-to-21.53× energy savings and 2.45-to-15.45× area reduction.

The attention engine is shown in Fig. 16.2.5, realizing the sparse local attention and QK sharing. Local-attention-based sparsity prunes weakly related distant Q/K pairs and preserves highly associated neighboring ones, thereby saving power consumption for the skipped $Q \bullet K^T$ and $P \bullet V$ computing. Between two adjacent local attention spans, only one K is different. Hence, a sliding-span scheme is mapped onto CIM macros, where the pop-out K_0 is directly updated by the push-in K_6, while K_1-K_5 are stationary. During computing, only one K_j is selected to multiply with Q_i according to the CIM address decoder. By changing the address range from K_j to K_{j+L-1}, the macro realizes a configurable local attention span size L. Applied to Enwik-8 and Text-8 tasks, the engine achieves 4.42×/4.20× energy savings and 2.71×/2.51× CIM size reduction. In addition, the engine also supports QK sharing – an efficient transformer algorithm to eliminate hardware overhead referring to K, which can be obtained as $K_j = Q_j / |Q_i|$. In such case, the weight storage and computing for the K generation can be eliminated. Also, $Q \bullet K^T$ becomes a diagonal-symmetric matrix, where half of the inner products can be reused with a different scaling factor $|Q_i|^{-1}$. Two external buffers, one for previously calculated inner products $Q_i \bullet Q_j$ and the other for the Q module $|Q_i|^{-1}$ are equipped for reuse. With both local attention and QK sharing, the engine can further realize 6.37×/5.99× energy savings and 3.61×/3.26× CIM memory reduction.

A prototype chip is fabricated in 28nm CMOS. The die photo and performance summary are given in Fig. 16.2.7. Fig. 16.2.6 gives the measurement results. To evaluate both macro and system performance, power domains for custom macros and other synthesized digital circuits are separated. The working frequency of the chip is up to 320MHz, while the overall power consumption is 250.65mW. Evaluation is conducted on a 12-layer 8-head Adaptive-Attention-Span transformer model with 1024 input tokens for the Enwik-8 and Text-8 tasks. A casual mask is applied to the results of $Q \bullet K^T$ to prevent it from attending to the further token. The FFM achieves an energy efficiency of 15.60TOPS/W at 20MHz and 0.60V, among which the IMPERA macros consume 41.48% of the power. The peak energy efficiency of AM is 70.37TOPS/W at 40MHz and 0.67V. The chip achieves 11.83/53.83/25.22TOPS/W system energy efficiency for $Q/K/V$ generation, $Q \bullet K / P \bullet V$ computing and overall attention, respectively. Compared to FP32 dense attention, the digital CIM accelerator causes 0.026/0.151BPC loss on Enwik-8/Text-8. The chip occupies 3.93mm^2, resulting in a 0.85TOPS/mm^2 area efficiency. In contrast with other SOTA designs, the accelerator exploits the CIM sparsity-aware architecture for both unstructured weight pruning and local attention. Although the IMPERA macro takes up 42.16% of the area of the total CIM (FFM+AM), the IMPERA-based FFM macro reduces the weight storage and multiply-accumulate circuit complexity for $Q/K/V$ generation. Meanwhile, the AM macro reduces the attention score storage and processing latency for $Q \bullet K / P \bullet V$ computation. These features facilitate efficient transformer acceleration, leading to 1.23-to-3.65× energy efficiency and 1.42-to-42.50× area efficiency improvement.

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References:

[1] J. Yue et al., "STICKER-IM: A 65 nm Computing-in-Memory NN Processor Using Block-Wise Sparsity Optimization and Inter/Intra-Macro Data Reuse," *IEEE JSSC*, vol. 57, no. 8, pp. 2560-2573, 2022.
[2] F. Tu et al., "A 28nm 15.59μJ/Token Full-Digital Bitline-Transpose CIM-Based Sparse Transformer Accelerator with Pipeline/Parallel Reconfigurable Modes," *ISSCC*, pp. 466-467, 2022.
[3] W. Niu et al., "PatDNN: Achieving Real-Time DNN Execution on Mobile Devices with Pattern-based Weight Pruning," *ACM ASPLOS*, pp. 907–922, 2020.
[4] N. Kitaev et al., "Reformer: The Efficient Transformer," *ICLR*, 2020.
[5] T. Tambe et al., "A 16-nm SoC for Noise-Robust Speech and NLP Edge AI Inference With Bayesian Sound Source Separation and Attention-Based DNNs," *IEEE JSSC*, 2022.
[6] Y. Wang et al., "A 28nm 27.5TOPS/W Approximate-Computing-Based Transformer Processor with Asymptotic Sparsity Speculating and Out-of-Order Computing," *ISSCC*, pp. 464-465, 2022.

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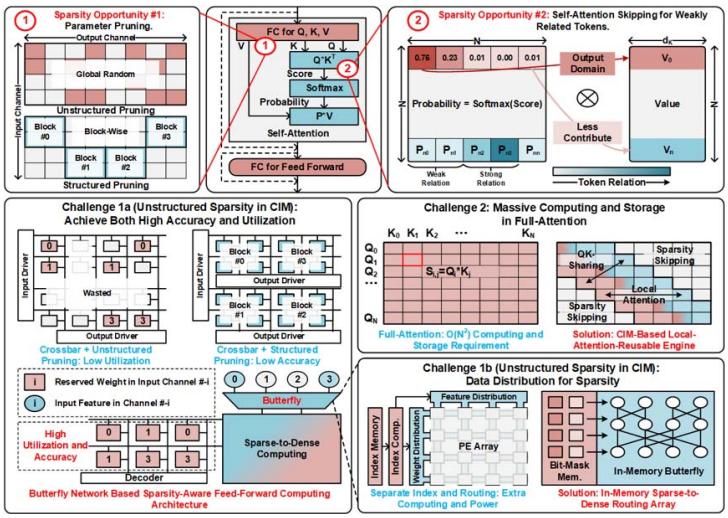


Figure 16.2.1: Challenges raised by transformer model sparsity and the proposed solution.

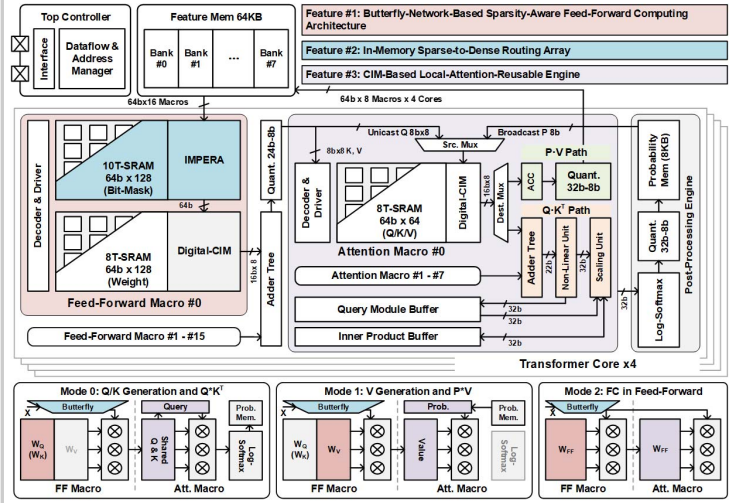


Figure 16.2.2: Overall architecture and operation modes of the accelerator.

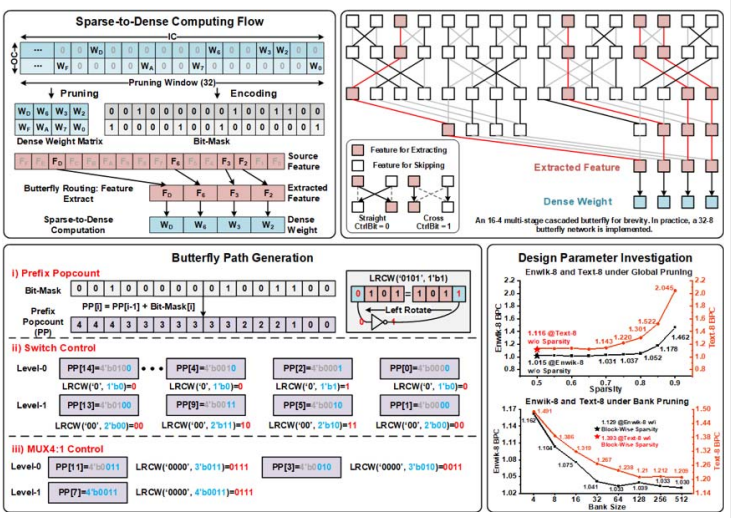


Figure 16.2.3: Butterfly-network-based sparsity-aware feed-forward zero-skipping scheme.

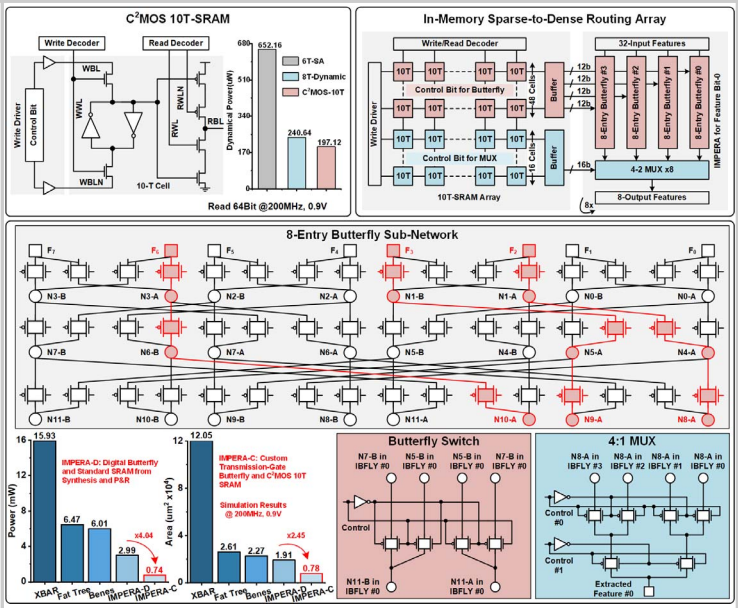


Figure 16.2.4: Circuit design of the in-memory sparse-to-dense routing.

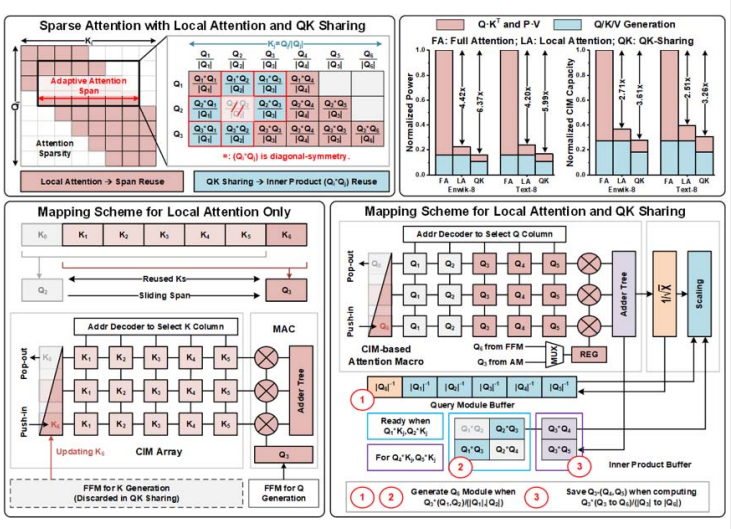
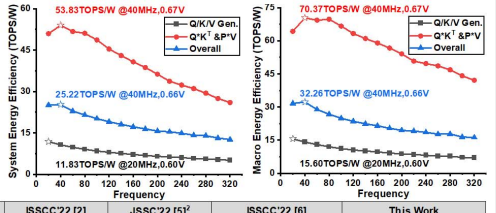


Figure 16.2.5: CIM based local-attention-reusable engine supporting variable attention span and QK sharing.

Evaluation on Different Tasks[1]

Model	Adaptive-Span Transformer	
Data Sets	Enwik-8	Text-8
Average Attention Window	83	98
BPC	1.041	1.267
BPC Loss[2]	-0.026	-0.151
Latency[3,4]	841ms	895ms
Overall Macro Energy Efficiency (TOPS/W)[3,5]	32.17	28.76
System Energy Efficiency (TOPS/W)[3,5]	25.22	22.82

1. Off-chip memory access is excluded;
2. Compared with dense attention in 32-b floating point;
3. Token number = 1024;
4. @320MHz, 1.03V; 5. @40MHz, 0.67V.

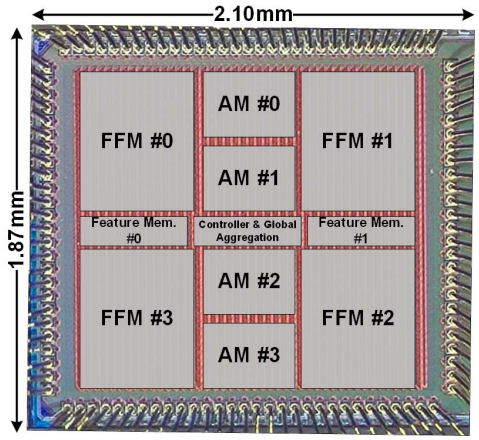


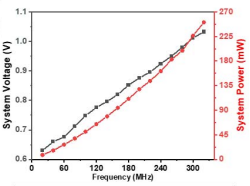
	JSSC'22 [1][1]	ISSCC'22 [2]	JSSC'22 [5][2]	ISSCC'22 [6]	This Work
Technology	65nm	28nm	16nm	28nm	**28nm**
Implementation	Analog CIM	Digital CIM	Digital	Digital Approximation	**Digital CIM**
Function	CNN	Transformer	Attention RNN	Transformer	**Transformer**
Precision	INT4	INT8/16	FP8	INT12	**INT8**
Die Area(mm²)	9.00	6.83	8.84	6.82	**3.93**
Supply Voltage (V)	0.90 - 1.05	0.60 - 1.00	0.55 - 1.00	0.56 - 1.10	**0.64 - 1.03**
Frequency (MHz)	50 - 100	80 - 240	130 - 573	50 - 510	**20 - 320**
Power (mW)	31.8 - 65.2	27.04 - 118.21	214	12.06 - 272.80	**8.27 - 250.65**
Peak Performance (TOPS or TFLOPS)	0.21	1.48	1.17	0.52 - 4.07[3]	**0.49 - 3.33**[5,6]
Macro Energy Efficiency (TOPS/W)	30.62	/	/	/	**FFM: 1.76 - 15.60**[5] **AM: 3.41 - 70.37**[6]
System Energy Efficiency (TOPS/W)	6.91	Overall[4]: 20.5	2.6 - 7.8	Q/K/V Gen.: 1.91 - 4.25 Q·K^T & P·V: 2.60 - 27.56[3]	**Q/K/V Gen.: 11.83**[5] **Q·K^T & P·V: 53.83**[6] **Overall: 25.22**[4,5,6]
Peak Area Efficiency (TOPS/mm2)	0.02	0.22	0.13	0.08 - 0.60[3]	**0.85**[5,6]

1. Performance under INT4 feature and weight; 2. FlexASR circuit performance for attention RNN;
3. 90% output sparsity ratio in P and V multiplication; 4. Energy efficiency for the entire attention;
5. 75% sparsity in weight for Enwik-8 task; 6. 92% sparsity in attention span for Enwik-8 task. Token Number = 1024.

Figure 16.2.6: Measurement results and performance comparison table.

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	Specifications	
Technology	28nm	
Precision	INT8	
CIM Size	8T	80KB
	10T	64KB
Digital SRAM	96KB	
Die Area(mm^2)	3.93	
Voltage (V)	0.64 - 1.03	
Frequency (MHz)	20 - 320	
Power (mW)	8.27 - 250.65	
Latency (ms)	Enwik8	841
	Text8	896
Perk Performance (TOPS)	3.33 @320MHz, 1.03V	
Energy Efficiency (TOPS/W)	Overall Macro	2.52 - 32.17
	System	1.96 - 25.22
Area Efficiency (TOPS/mm^2)	0.85	

Figure 16.2.7: Die micrograph, voltage-frequency scaling and chip summary table.

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